

COE231: Human Computer Interaction

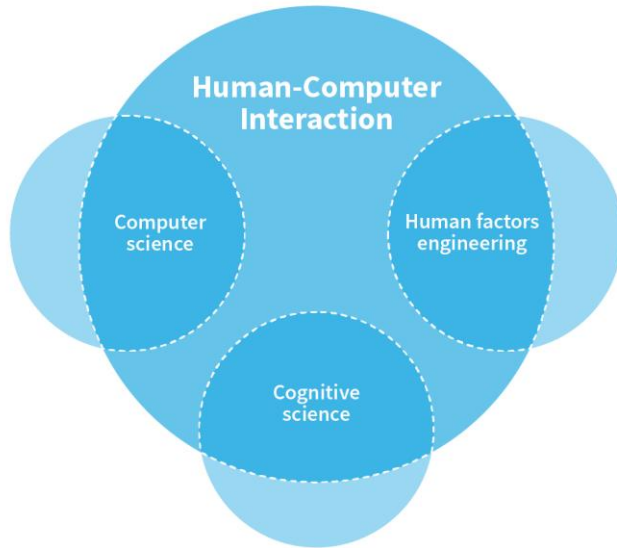
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Human Computer Interaction

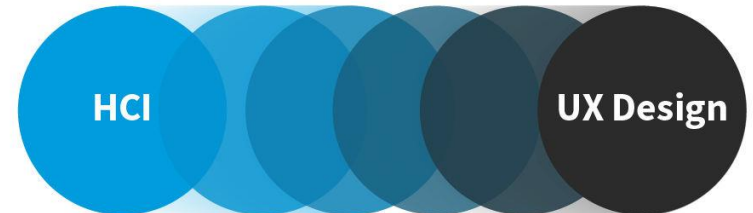
- **Human-computer interaction (HCI)**: is a multidisciplinary field of study that focuses on the design of computer technology, particularly the interaction between humans (users) and computers.

The Multidisciplinary Field of HCI



Interaction Design Foundation
interaction-design.org

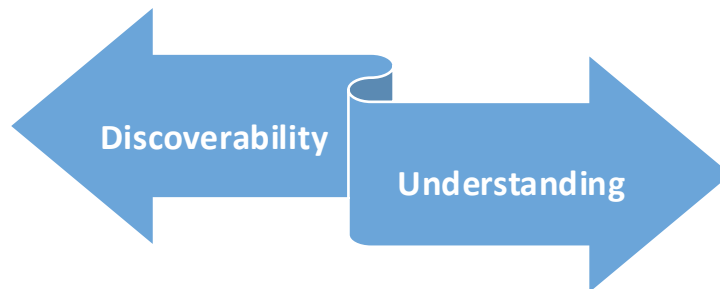
The Evolution of UX Design



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Psychopathology of Everyday Things

- 'Norman doors'
 - confusing door designs.
- Good design
 - make **operations intuitive** without requiring signs or instructions.
- Introduces two essential design principles:
 - **Discoverability:**
 - Can users understand what actions are possible?
 - **Understanding:**
 - Can users interpret the product's purpose and function correctly?



Psychopathology of Everyday Things

- Poor design leads to frustration and inefficiency
- Suggested solutions for better design
 - Add **visible cues** like vertical plates to indicate where to push or pull.
 - Ensure functional components are **obvious** and **naturally interpreted**.



Complexity and Usability

- Increasing complexity of modern devices
 - Stoves to washing machines.
 - Example: An Italian washer-dryer had so many functions that users only memorized one setting.
- Root of complexity issues:
 - Designers' focus on technical capabilities rather than usability.
 - Lack of understanding of user psychology.



Three Types of Design

- Role of industrial design, interaction design, and experience design
 - Industrial design focuses on form and material.
 - Interaction design emphasizes usability and understanding.
 - Experience design considers emotional impact and enjoyment.

Three Types of Design

- Industrial Design (Form and Material)
 - Example: Apple MacBook
 - sleek aluminum body,
 - lightweight design, and
 - precision engineering reflect industrial design principles.
 - material choices
 - durability,
 - aesthetics, and
 - a premium feel,
 - enhances both function and visual appeal.



Three Types of Design

- Interaction Design (Usability and Understanding)
- Example: Google Search Interface
 - simplicity of homepage,
 - clear search bar
 - intuitive auto-suggestions
- Easily understand how to input queries and receive relevant results without confusion.



Google Search

I'm Feeling Lucky

Three Types of Design

- Experience Design (Emotional Impact and Enjoyment):
- Example: Disneyland Theme Parks
 - immersive experience through
 - themed environments,
 - music, and
 - interactive attractions.
 - The park's design evokes joy and nostalgia,
 - creating emotional connection for visitors.



Machines should be
designed to align with
human behavior and
limitations, not the
other way around!

Human-Centered Design (HCD)

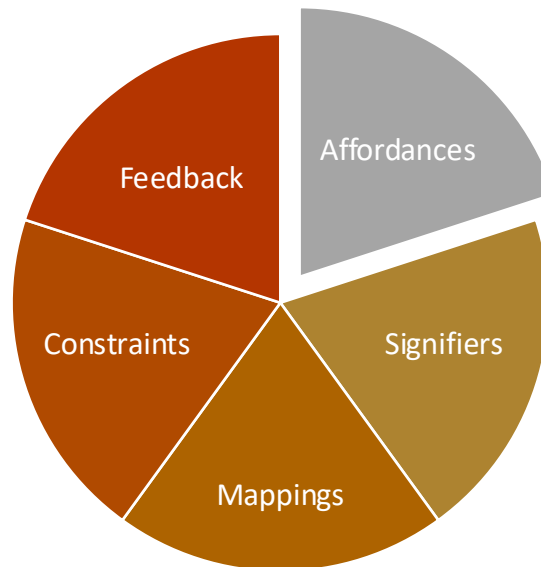
- HCD
 - human needs and capabilities first in the design process.
- Good design
 - effective communication between the machine and the user.
- Challenges:
 - Rapid technological advancements often outpace improvements in design.
 - New technologies often repeat old design mistakes.
- Iterative design
 - testing, refining, and adjusting based on real-world use.
- Designers should prioritize **error scenarios** and ensure the design communicates solutions.

Enhancing 'Discoverability'

Fundamental Principles of Interaction

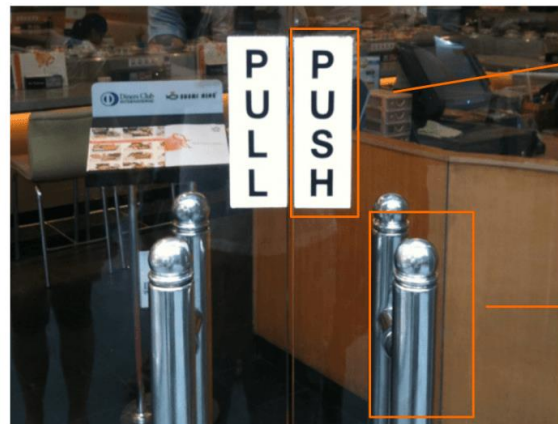
- Five psychological principles

- Affordances: Relationships between objects and users that suggest possible actions.
- Signifiers: Indicators that communicate where and how to act.
- Mappings: Clear relationships between controls and their effects.
- Constraints: Limitations that guide users toward correct usage.
- Feedback: Information that confirms actions and their results.



Fundamental Principles of Interaction

- Affordances and signifiers are particularly crucial
 - Affordances determine what actions are possible.
 - Signifiers indicate where and how actions should take place.
- Examples
 - A flat plate on a door suggests pushing.
 - Birds mistakenly fly into glass windows due to lack of visible anti-affordance signals.



Signifier
- Sign to explain what to do (because the affordances are confusing)

Affordance
- Handle to physically grasp

Affordances

- Possible interactions between a user and an object, determined by their physical properties.
 - Examples of affordances:
 - A chair affords sitting and lifting.
 - A door handle affords pulling or turning.
- Depend on the relationship between the object's design and the user's capabilities.
- When affordances are invisible or misleading
 - Example: Glass windows afford seeing through but not passing through, which may confuse birds or people.



Affordances

- affordances describe possibilities
 - their **visibility** is essential for effective design.
- Misinterpreted or hidden affordances can lead to errors and frustration.
- Designers must ensure affordances are perceivable and align with user expectations.



Signifiers

- Designers often confuse affordances and signifiers
 - Example: A circle on a screen suggests touch interaction but is technically a signifier, not an affordance.
- Signifiers provide essential clues for understanding and using products effectively.
- Examples:

Case	Good or Bad?
A stop sign with a red octagonal shape and bold white letters.	
USB ports on a device without proper labeling or color coding.	
App icons that visually represent their functions (e.g., a camera icon for the camera app)	
A door without any visible handle, push plate, or labeling.	

Signifiers

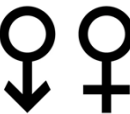
- Deliberate vs. accidental signifiers:
 - A push label on a door is deliberate.
 - Trails in snow indicating a path are accidental.
- Good design integrates signifiers seamlessly into the user experience.
- Key takeaway: **Signifiers are more important than affordances in communicating usability.**

Example 1: Smartphone Touchscreen Keyboard

A smartphone touchscreen keyboard relies heavily on **signifiers** rather than **affordances** to communicate usability.

Example 2: Airport Restroom Signs

In an airport, restroom doors afford entry, but without proper **signifiers**, such as universally recognized male and female icons, users might not know which restroom to enter.



A.



B.



C.



A.



B.



C.



A.



B.



C.



D.



FIGURE 1.5. Accidental Affordances Can Become Strong Signifiers. This wall, at the Industrial Design department of KAIST, in Korea, provides an anti-affordance, preventing people from falling down the stair shaft. Its top is flat, an accidental by-product of the design. But flat surfaces afford support, and as soon as one person discovers it can be used to dispose of empty drink containers, the discarded container becomes a signifier, telling others that it is permissible to discard their items there. (Photographs by the author.)

A.



B.



C.





FIGURE 1.6. Signifiers on a Touch Screen. The arrows and icons are signifiers: they provide signals about the permissible operations for this restaurant guide. Swiping left or right brings up new restaurant recommendations. Swiping up reveals the menu for the restaurant being displayed; swiping down, friends who recommend the restaurant.

Mapping

- Mapping: relationship between controls and their effects.
- Good mapping is intuitive and spatially logical.
 - Example: A car seat adjustment control shaped like the seat makes its function clear.



FIGURE 1.7. Good Mapping: Automobile Seat Adjustment Control. This is an excellent example of natural mapping. The control is in the shape of the seat itself: the mapping is straightforward. To move the front edge of the seat higher, lift up on the front part of the button. To make the seat back recline, move the button back. The same principle could be applied to much more common objects. This particular control is from Mercedes-Benz, but this form of mapping is now used by many automobile companies. (Photograph by the author.)

Mapping and Feedback

- Feedback: Essential for confirming actions.
 - Should be
 - immediate,
 - informative, and
 - appropriately prioritized.
 - Poor feedback causes confusion or annoyance (e.g., excessive beeping from appliances).
 - Overloading users with feedback can lead to important signals being ignored.

Enhancing 'Understanding'

Conceptual Models and System Image

- Conceptual models help users understand how a product works.
 - Example: Files and folders on computers are a simplified conceptual model.
- Misleading conceptual models can lead to frustration
 - a poorly designed refrigerator with interconnected controls.



Hotel Room Light Switches with Keycard Holders

In many hotels, inserting a keycard into a holder activates the room's electrical system. Guests unfamiliar with this setup may assume the lights are broken when flipping traditional switches doesn't work. This leads to unnecessary frustration and calls to the front desk for assistance.

Conceptual Models and System Image

- The "system image" is the total information available to users about a product
- It is derived from
 - Design and layout.
 - Manuals, labels, and online resources.
- Clear communication through the system image is vital for usability.

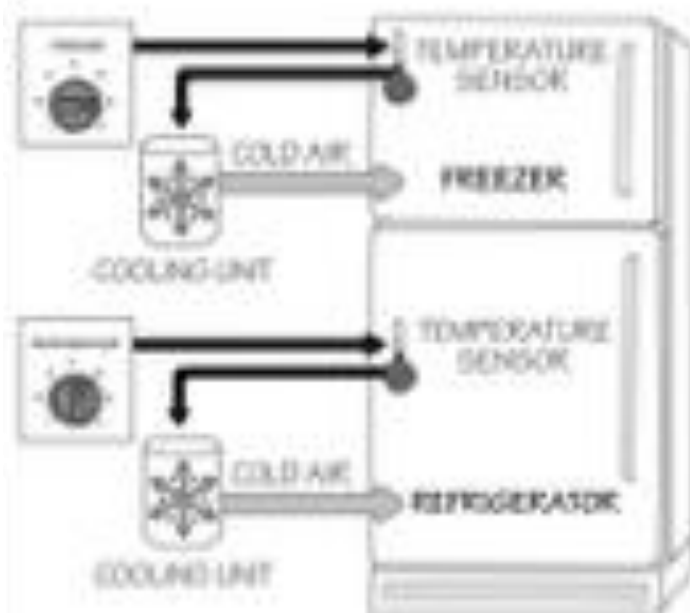


FIGURE 1.8. Junghans Mega 1000 Digital Radio Controlled Watch. There is no good conceptual model for understanding the operation of my watch. It has five buttons with no hints as to what each one does. And yes, the buttons do different things in their different modes. But it is a very nice-looking watch, and always has the exact time because it checks official radio time stations. (The top row of the display is the date: Wednesday, February 20, the eighth week of the year.) (Photograph by the author.)



FIGURE 1.9. Refrigerator Controls. Two compartments—fresh food and freezer—and two controls (in the fresh food unit). Your task: Suppose the freezer is too cold, the fresh food section just right. How would you adjust the controls so as to make the freezer warmer and keep the fresh food the same? (Photograph by the author.)

A.



B.

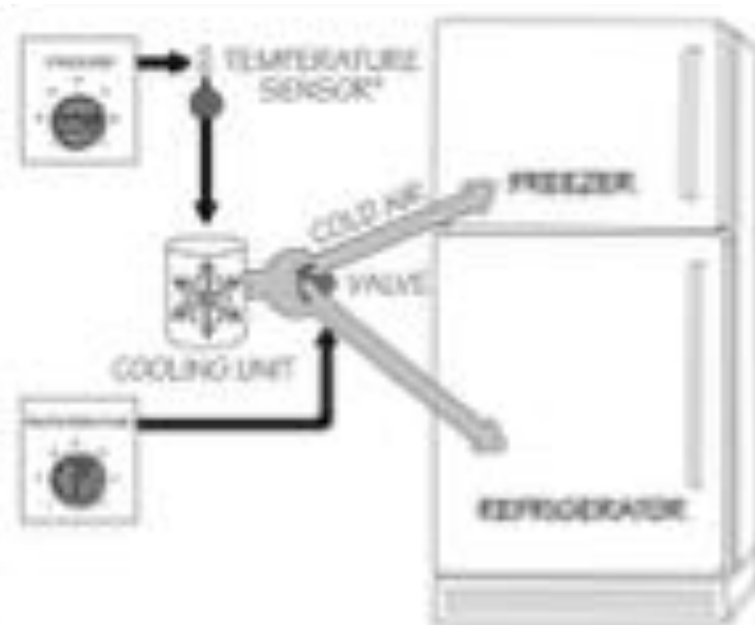


FIGURE 1.11. The Designer's Model, the User's Model, and the System Image. The designer's conceptual model is the designer's conception of the look, feel, and operation of a product. The system image is what can be derived from the physical structure that has been built (including documentation). The user's mental model is developed through interaction with the product and the system image. Designers expect the user's model to be identical to their own, but because they cannot communicate directly with the user, the burden of communication is with the system image.

